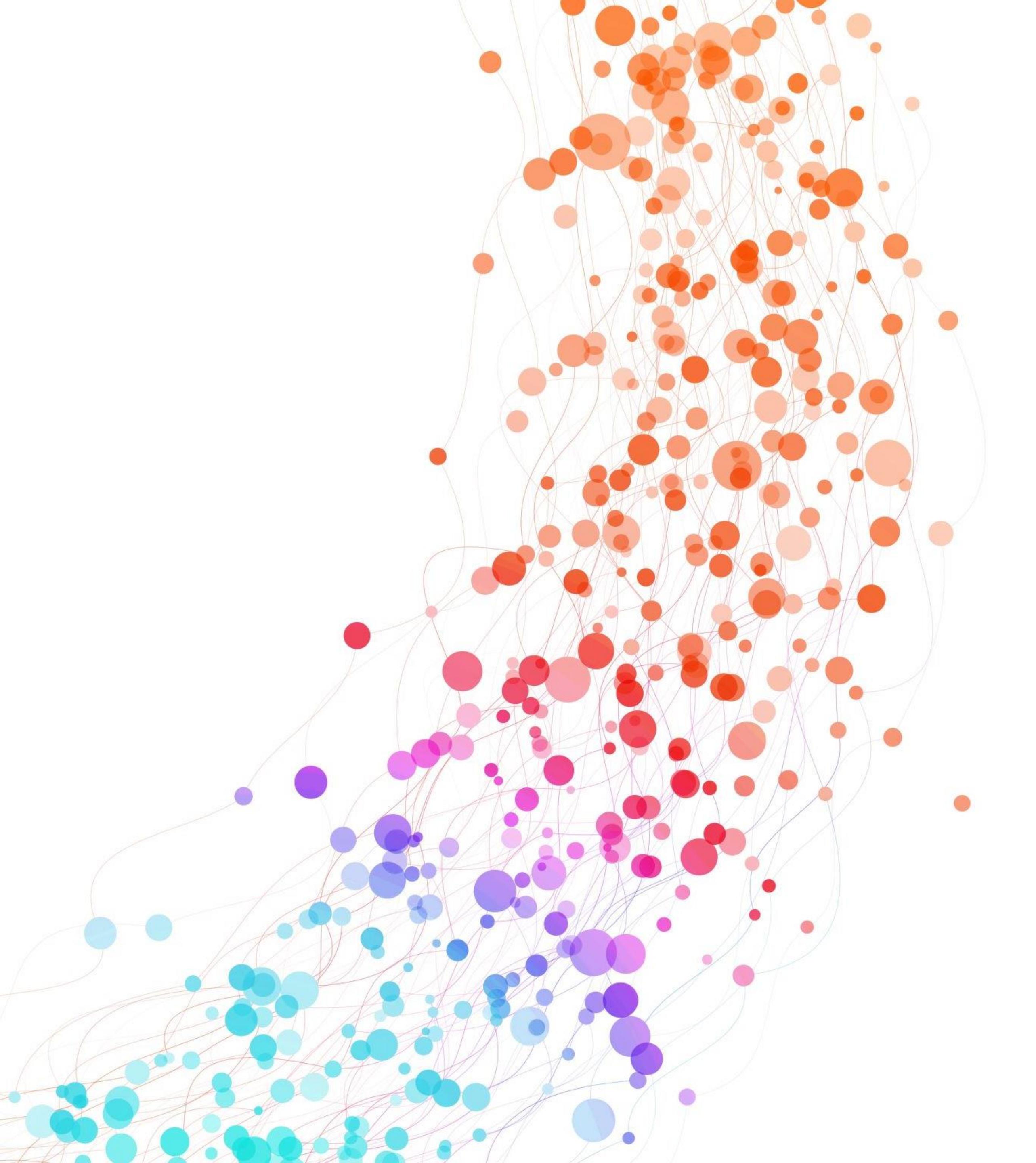
Geospatial Database Using MongoDB

everaging MongoDB for Geospatial Data Management

Atta Muhammad

92333197110



Introduction to Geospatial Databases

Definition: Databases designed to store, query, and manipulate spatial data.

Use Cases:

- Location-based services (e.g., Uber, Google Maps)
- Environmental monitoring
- Urban planning

Why MongoDB?

- NoSQL flexibility
- Built-in geospatial querying capabilities

MongoDB Geospatial Features

Support for GeoJSON objects:

- Point
- Line
- String
- Polygon

Index Types:

- 2d index
- 2dsphere index

Geospatial Query Operators:

- $geoWithin
- $near
- $geoIntersects

Sample MongoDB Geospatial Dataset

```json
{
  "name": "Pizza Palace",
  "type": "Restaurant",
  "location": {
    "type": "Point",
    "coordinates": [-73.856077, 40.848447]
  }
}
```

Creating the Sample Database

```
use geospatialDB;
db.createCollection("places");
```

Inserting a sample document

```
db.places.insertMany([
  {
    "name": "Pizza Palace",
    "type": "Restaurant",
    "location": {
      "type": "Point",
      "coordinates": [-73.856077, 40.848447]
    }
  },
  {
    "name": "Burger Haven",
    "type": "Restaurant",
    "location": {
      "type": "Point",
      "coordinates": [-73.856174, 40.849102]
    }
  }
])
```

Creating a Geospatial Index

```
geospatialDB > db.places.createIndex({ "location": "2dsphere" });
```

Querying Geospatial Data - $near

```
geospatialDB> db.places.find({
                  location: {
                    $near: {
                      $geometry: {
                        type: "Point",
                        coordinates: [-73.856077, 40.848447]
                      },
                      $maxDistance: 1000
                    }
                  }
                });
```

Querying Geospatial Data – $geoWithin

```
db.places.find({
  location: {
    $geoWithin: {
      $geometry: {
        type: "Polygon",
        coordinates: [
          [
            [-73.856174, 40.849102],
            [-73.855, 40.848],
            [-73.857, 40.847],
            [-73.856174, 40.849102]
          ]
        ]
      }
    }
  }
});
```

Querying Geospatial Data – $geoIntersects

Find places intersecting a specific geometry (e.g., a road).

```
geospatialDB> db.places.find({
                location: {
                  $geoIntersects: {
                    $geometry: {
                      type: "LineString",
                      coordinates: [
                        [-73.857, 40.847],
                        [-73.856, 40.849]
                      ]
                    }
                  }
                }
              });
```

Updating Geospatial Data

>_MONGOSH

```
geospatialDB> db.places.updateOne(
                { "name": "Pizza Palace" },
                { $set: { "location.coordinates": [-73.857077, 40.849447] } }
              );
```

Real-World Applications

Logistics: Optimize delivery routes.

Real Estate: Identify properties in specific areas.

Emergency Services: Locate nearest hospitals or police stations.

Advantages of Using MongoDB for Geospatial Data

Flexible schema design.

Scalable and distributed architecture.

Built-in support for geospatial data and queries.

Quetions